SESSION 4

1.

Insert at least 5 documents into a MongoDB collection called playlists, where each document represents a Spotify playlist with fields: name (string), likes (number), genre (string), and followers (number).

```
>_MONGOSH
test> db.playlists.insertMany([
        {
          name: "Top Pop Hits",
          likes: 1500,
          genre: "Pop",
          followers: 8000
        },
        {
          name: "Hip-Hop Vibes",
          likes: 2200,
          genre: "Hip-Hop",
          followers: 6500
        },
        {
          name: "Rock Legends",
          likes: 900,
          genre: "Rock",
          followers: 1200
        },
        {
          name: "Chill Beats",
          likes: 700,
```

2.

Write a MongoDB query using the $gt operator to find all playlists with more than 1000 followers in the playlists collection and display only the name and followers fields.

Hint: Use projection to limit the fields returned.

```
}
> db.playlists.find(
    { followers: { $gt: 1000 } },
    { _id: 0, name: 1, followers: 1 }
  );
< {
    name: 'Top Pop Hits',
    followers: 8000
  }
  {
    name: 'Hip-Hop Vibes',
    followers: 6500
  }
  {
    name: 'Rock Legends',
    followers: 1200
  }
  {
    name: 'Workout Mix',
    followers: 3000
  }
```

3.

Use the $in operator to find all playlists where the genre is either 'Pop' or 'Hip-Hop' and sort the results by likes in descending order.

```
> db.playlists.find(
    { genre: { $in: ["Pop", "Hip-Hop"] } }
  ).sort({ likes: -1 });
< {
    _id: ObjectId('6a3cbfb7fc9abc54bc8b749e'),
    name: 'Hip-Hop Vibes',
    likes: 2200,
    genre: 'Hip-Hop',
    followers: 6500
  }
  {
    _id: ObjectId('6a3cbfb7fc9abc54bc8b749d'),
    name: 'Top Pop Hits',
    likes: 1500,
    genre: 'Pop',
    followers: 8000
  }
```

4.

Query the playlists collection to find playlists with likes between 500 and 2000 (inclusive) using $gte and $lte, then limit the output to the top 3 playlists sorted by followers.

```
{
  _id: ObjectId('6a3cbfb7fc9abc54bc8b749d'),
  name: 'Top Pop Hits',
  likes: 1500,
  genre: 'Pop',
  followers: 8000
}
{
  _id: ObjectId('6a3cbfb7fc9abc54bc8b74a1'),
  name: 'Workout Mix',
  likes: 1800,
  genre: 'EDM',
  followers: 3000
}
{
  _id: ObjectId('6a3cbfb7fc9abc54bc8b749f'),
  name: 'Rock Legends',
  likes: 900,
  genre: 'Rock',
  followers: 1200
```

```
db.playlists.find(
  {
    likes: {
      $gte: 500,
      $lte: 2000
    }
  }
)
.sort({ followers: -1 })
.limit(3);
```

5.

Suppose you want to recommend trending playlists like Spotify does. Write a MongoDB query that finds all playlists with more than 5000 followers and at least 1000 likes, but only show their name and genre. Explain in one line how you combined multiple query operators in your filter.

```
spotifyDB> db.playlists.find(
             {
               followers: { $gt: 5000 },
               likes: { $gte: 1000 }
             },
             {
               _id: 0,
               name: 1,
               genre: 1
             }
           );
```

```
< {
    name: 'Top Pop Hits',
    genre: 'Pop'
  }
  {
    name: 'Hip-Hop Vibes',
    genre: 'Hip-Hop'
  }
```